

MENSTRUAL HEALTH TRACKER APPLICATION HERAURA

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ABSTRACT

HER AURA is an end-to-end mobile app, developed using Flutter and utilizing Firebase, that aims to offer personalized menstrual and reproductive health care. In response to the necessity of precise tracking, symptom relief, and individualized nutritional support, the app features an UI/UX developed in Figma, which makes it easy to log menstrual cycle information such as start and finish dates, duration of the cycle, and variety of symptoms ranging from cramps, tiredness, and mood swings. Advanced tracking makes the identification of trends and patterns simple, which aids users in greater understanding of their bodies. One of the essential features of HER AURA is its intelligent symptom-based food suggesting system. It evaluates logged symptoms and gives custom dietary recommendations to help relieve pain. For example, users suffering from extreme cramps can get suggestions for foods rich in magnesium, whereas people struggling with exhaustion can be encouraged to have meals rich in iron and energy-nourishing substances. The system combines a list of food categories and their nutrient content and correlates them with general menstrual symptoms. In addition, HER AURA provides assistance to pregnant users by providing trimester-specific nutritional advice, maximizing maternal and fetal health at each stage of pregnancy. This encompasses advice on required nutrients, portion control, and meal planning. The app also offers predictive insights into future periods, ovulation, and fertile windows, allowing users to efficiently plan their routine and make the right decisions on family planning. Prioritizing data privacy and security, HER AURA leverages Firebase's security features to safeguard sensitive health data. The intuitive UI/UX design of the app makes it easy to navigate and access core features, and integrated educational resources such as articles and links to credible sources inform users about menstrual and pregnancy health.

1. INTRODUCTION

Menstrual health tracking is essential for individuals to understand their reproductive health, manage symptoms, and receive personalized wellness recommendations. The **HER AURA** Menstrual Tracker is designed to help users monitor their cycles, track symptoms like cramps, fatigue, and mood swings, and receive tailored food suggestions to alleviate discomfort. Additionally, the app provides pregnancy-week-specific nutritional guidance to support maternal health.

Users can log their period details, track symptoms, and receive cycle predictions, helping them plan their routines effectively. The app integrates reminders for hydration, medication, and upcoming periods, ensuring users stay informed about their health. HER AURA's unique food recommendation system suggests diet plans based on logged symptoms. For example, if a user experiences severe cramps, the app may recommend anti-inflammatory foods rich in magnesium and omega-3 fatty acids. Similarly, for fatigue, it suggests iron-rich leafy greens and vitamin C sources.

The pregnancy-related nutrition feature ensures expecting mothers receive optimal dietary guidance during each trimester. Built with **Flutter** for front-end development, **Firebase** for secure data storage, and **Figma** for UI/UX design, HER AURA ensures a smooth user experience with an intuitive interface.

The app's symptom tracking feature enables users to record changes such as bloating, nausea, and headaches, helping them identify patterns over time and take proactive measures. The app's user-friendly dashboard allows easy navigation, customizable settings, and personalized cycle predictions. Integration with Android Studio Emulator ensures compatibility across devices, making it accessible to a wider audience.

2. LITERATURE REVIEW

AI-Powered Menstrual Cycle Prediction for Reproductive Health - Menstrual tracking apps play a crucial role in reproductive health by providing users with personalized cycle predictions and symptom tracking [1].

Enhancing Ovulation Forecasting with AI Models - Research highlights the effectiveness of AI-driven predictions in improving cycle accuracy and ovulation forecasting [2].

User Engagement Strategies in Menstrual Health Apps - Studies emphasize the importance of user-friendly interfaces and engagement strategies to enhance adherence to menstrual tracking [3].

Nutritional Recommendations for Menstrual Health Management - The integration of food recommendations based on symptoms has been shown to positively impact menstrual health and reduce discomfort [4].

Mobile Health Applications for Hormonal and Emotional Well-being - Mobile health (mHealth) applications incorporating real-time analytics offer better insights into hormonal fluctuations and emotional well-being [5].

Cloud-Based Data Storage and Security in Health App - The adoption of Firebase and cloud-based storage enhances data security and accessibility for menstrual health apps [6].

Cross-Platform Development: React Native vs. Flutter in mHealth Apps - Research suggests that React Native and Flutter frameworks enable cross-platform compatibility, improving usability and outreach [7].

Ethical Considerations and Privacy Concerns in Menstrual Apps - Privacy concerns and ethical considerations in menstrual apps are critical, necessitating encrypted data handling and secure authentication mechanisms [8].

Impact of Personalized Notifications on Menstrual Health Awareness - Studies indicate that personalized notifications and reminders encourage consistent tracking and better menstrual health awareness [9].

AI-Powered Food Suggestions for Menstrual Symptom Management - The impact of hormonal changes on dietary needs has led to the development of AI-powered food recommendations tailored to different cycle phases [10].

Machine Learning in Predicting Irregular Menstrual Cycles - Machine learning models help predict irregular cycles, aiding users with conditions like PCOS and endometriosis [11].

Gamification for Improved User Engagement in Health Apps - Gamification elements, such as achievement badges and progress tracking, improve user motivation and engagement [12].

Integrating Mental Health Features in Menstrual Tracking Apps - Research supports the inclusion of mental health tracking features, as mood fluctuations are closely linked to hormonal changes [13].

Real-Time Data Synchronization in Menstrual Health Apps Using Firebase - The use of Firebase for real-time data synchronization enhances the performance and responsiveness of menstrual tracking apps [14].

Wearable Technology for Physiological Monitoring in Menstrual Tracking - The integration of wearable devices for physiological data collection is emerging as a promising feature for next-generation menstrual tracking solutions [15].

3. METHODOLOGY

The development of **HER AURA**, a menstrual health tracking application, follows a structured approach that integrates UI/UX design, front-end and back-end development, and database management to provide an intuitive and personalized experience for users. The methodology involves multiple stages, including system design, implementation, and testing.

A. UI/UX Design

The initial phase involved creating a **Figma-based prototype** to visualize the app's interface and user experience. This helped define the navigation flow, layout, and key features, ensuring an intuitive design that enhances user engagement.

B. Front-End Development

The mobile application was developed using **Flutter**, a cross-platform framework that enables smooth UI implementation and efficient performance on both Android and iOS devices. Flutter was chosen for its flexibility in creating dynamic and visually appealing interfaces.

C. Back-End and Database Management

To manage user data securely, **Firestore** was integrated as the primary back-end and database solution. Firestore Authentication was used for user login and registration, while Firestore Database stored menstrual cycle details, symptoms, and food suggestions. This cloud-based solution ensures seamless real-time data updates and synchronization across multiple devices.

D. Feature Implementation

- **Menstrual Cycle Tracking:** Users can log their period start and end dates, allowing the system to analyze cycle patterns and predict future dates.
- **Symptom Logging:** The app enables users to record symptoms such as cramps, fatigue, and mood changes.

- **Food Recommendations:** Based on logged symptoms, the app suggests foods that can help alleviate discomfort. Additionally, for pregnant users, **pregnancy-week-specific dietary recommendations** are provided to ensure proper nutrition.
- **Notification and Reminders:** The app sends personalized notifications for upcoming periods, medication reminders, and hydration alerts.

E. Testing and Optimization

The app underwent rigorous testing on the **Android Studio Emulator** and physical devices to check for UI responsiveness, database connectivity, and overall performance. User feedback was gathered for iterative improvements, ensuring accuracy in period predictions and personalized recommendations.

F. Data Privacy and Security

To protect user data, **Firebase Authentication** was used for secure login, and **encryption mechanisms** were implemented to prevent unauthorized access. The app follows strict data privacy protocols to ensure confidentiality.

This structured methodology ensures that HER AURA provides users with a reliable, secure, and user-friendly menstrual health tracking experience.

3.1 FLOWCHART

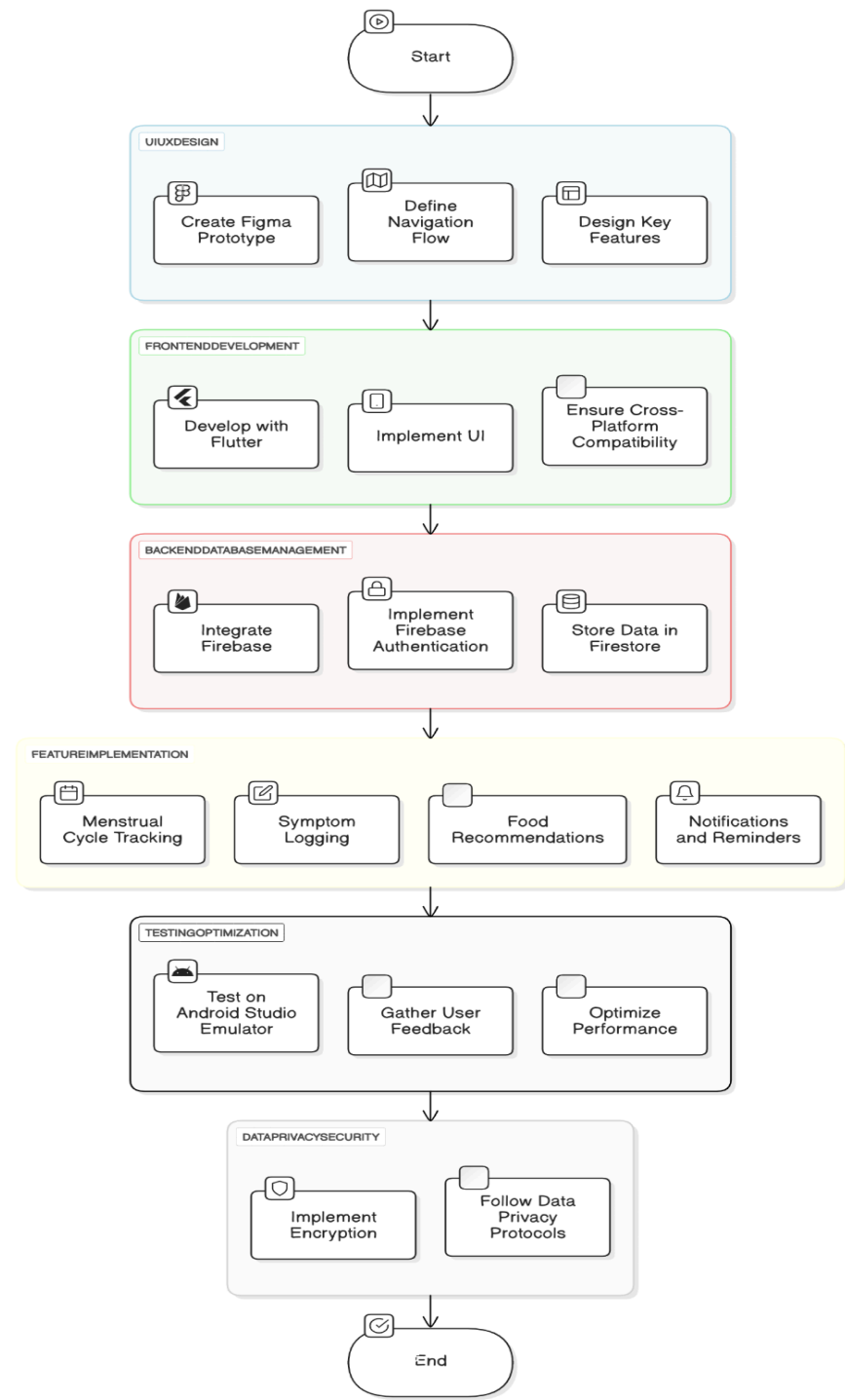


Figure 3.1. MENSTRUAL TRACKER FLOWCHART

3.2 USER INTERFACE DESIGN

The UI of some particular screens was designed using Figma before implementation in Flutter to better understand the colour grading and Figma is a collaborative interface design tool that connects everyone in the design process so teams can deliver better products, faster. (No-code Platform) Example : Splash Screen and Signup Screen.

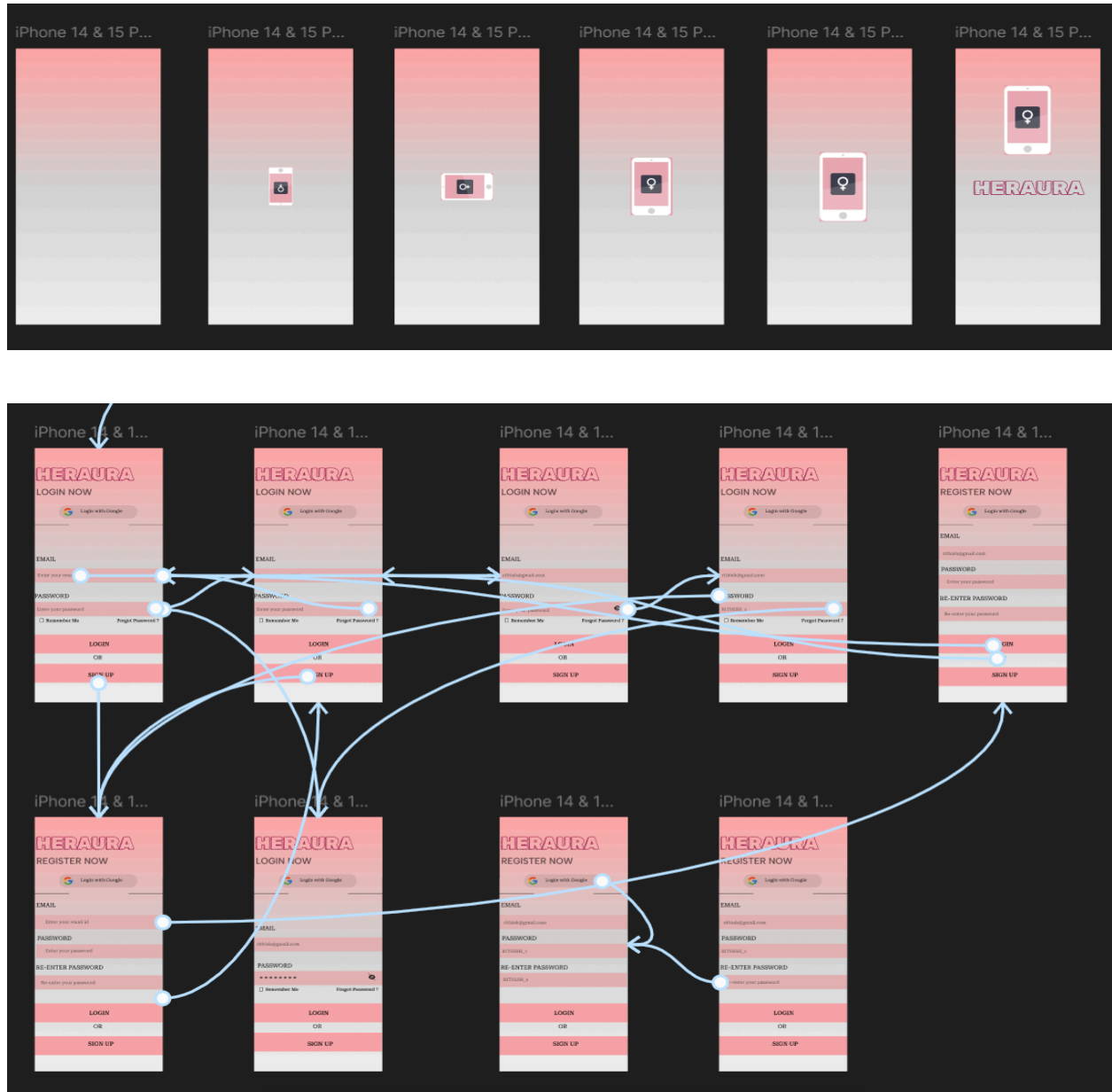


Figure 3.2. Figma Design of Splash and Login Screen

3.3 SYSTEM ARCHITECTURE

The HER AURA menstrual tracker follows a three-tier architecture, ensuring an intuitive user experience, real-time data updates, and secure storage.

The system consists of the front-end (UI layer), back-end (Firebase services), and database (Firestore for data storage).

3.3.1 Front-End (User Interface Layer)

- Developed using Flutter, enabling a cross-platform experience on Android and iOS.
- Designed in Figma to create a structured and visually appealing UI.
- Allows users to register, log their menstrual cycles, track symptoms, and receive food suggestions.
- Retrieves real-time data from Firestore and updates the UI with cycle insights, reminders, and nutritional recommendations.

3.3.2 Backend (Firebase Services)

- Firebase Authentication: Ensures secure login and personalized user experience.
- Cloud Functions (if used): Automates reminders and notifications.
- Firebase Firestore: Serves as a cloud database, storing user profiles, cycle logs, symptom data, and food recommendations.
- Firebase Cloud Messaging (FCM): Sends push notifications for cycle reminders, hydration alerts, and medication reminders.

3.3.3 Database (Storage Layer)

- Uses Firestore, a NoSQL database, for real-time data storage and retrieval.
 - User Profiles: Name, email, cycle details.
 - Menstrual Logs: Start and end dates, cycle patterns.
 - Symptom Data: Cramps, fatigue, mood changes.
 - Food Recommendations: Based on cycle phase and pregnancy week.

3.3.4 Additional Components

- Notification System – Uses Firebase Cloud Messaging to send cycle-related reminders.
- Security & Privacy – Implements data encryption and authentication measures to protect user health information.

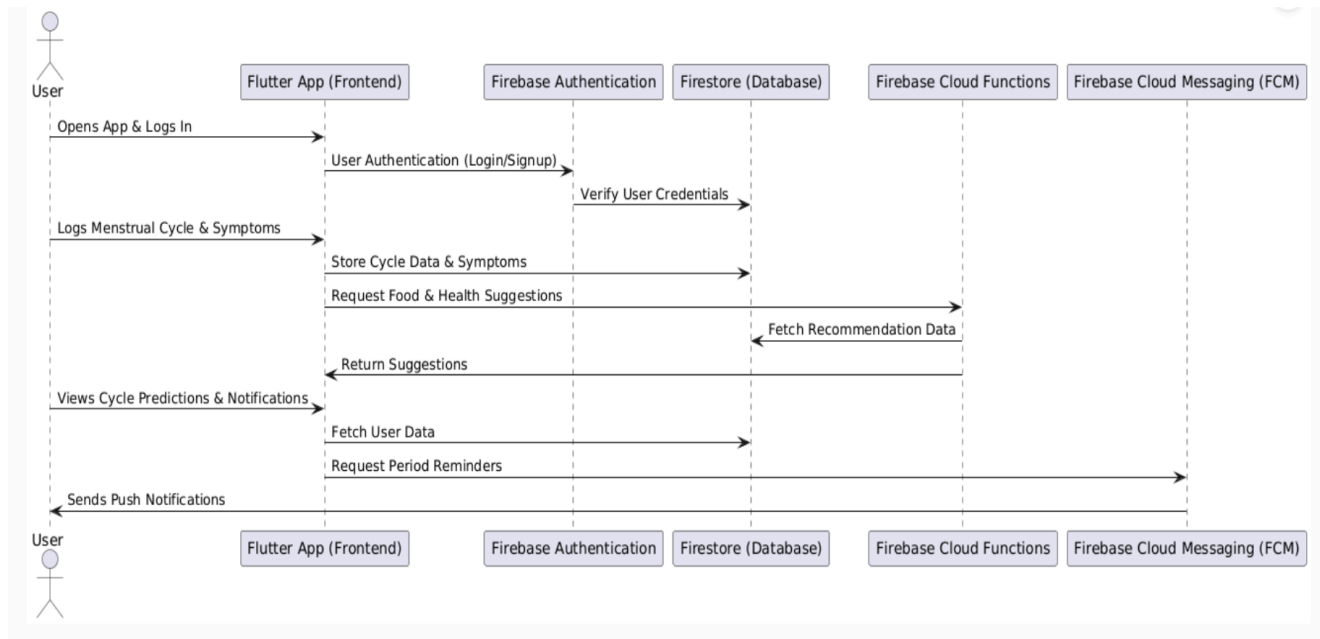


Figure 3.3. System Architecture

3.4 FIREBASE AUTHENTICATION

Firebase Authentication is used in the HER AURA menstrual tracker to securely manage user sign-ups and logins using email and password authentication. When a user registers, Firebase assigns a unique User ID (UID) and securely stores the password in an encrypted format rather than saving it as plain text. This ensures strong security for user credentials.

Once the user logs in, the app retrieves their menstrual cycle data from Firestore, Firebase's cloud-based NoSQL database. Firestore organizes user data into collections and documents, where each user has a unique document storing their:

- Email ID (for authentication)
- Cycle Start Date (first day of the period)
- Cycle Length (average duration of the cycle)
- Period Duration (number of days the period lasts)
- Symptoms logged (e.g., cramps, fatigue)

To ensure security, Firebase Authentication uses hashed passwords, authentication tokens, and Firestore Security Rules, allowing only the logged-in user to access and modify their data. Firestore's real-time synchronization updates the database instantly whenever the user logs a new symptom or modifies their cycle details.

This integration of Firebase Authentication with Firestore ensures data privacy, secure login, and a seamless user experience, making HER AURA a reliable menstrual health tracker.

heraura ▾

Authentication

Users Sign-in method Templates Usage Settings Extensions

i The following Authentication features will stop working when Firebase Dynamic Links shuts down on August 25, 2025: email link authentication for mobile apps, as well as Cordova OAuth support for web apps. ▾

🔍 Search by email address, phone number, or user UID

Add user



Identifier	Providers	Created ▾	Signed In	User UID
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		+ Add field
		createdAt: March 29, 2025 at 3:01:21 PM UTC+5:30
		cycleLength: 28
		email: "rithish123@gmail.com"
		firstname: "Rithish"
		lastPeriodStart: March 15, 2025 at 12:00:00 AM UTC+5:30
		lastname: "R"
		periodLength: 5
		uid: "4AWNqRblgxYomnB4wvP7M1IX0wY2"

Figure 3.4. FireBase Authentication

3.5 FLUTTER IMPLEMENTATION

Flutter, developed by Google, is a powerful open-source UI framework that allows for the development of high-performance cross-platform mobile applications. The HER AURA menstrual tracker app was built using **Flutter** to provide a smooth, responsive, and visually appealing user experience on both Android and iOS devices. The development process began with **Figma**, a popular UI/UX design tool, to visualize the interface before implementing it in Flutter.

The app consists of multiple interactive screens that allow users to **register, log in, track their menstrual cycle, record symptoms, receive food recommendations, and access pregnancy-related nutrition guidance**. The UI is built using Flutter's **widget-based architecture**, which provides modularity and flexibility. Widgets such as **TextFields, Buttons, Cards, ListViews, and Calendars** were used to create an engaging and intuitive interface.

One of the key challenges in developing the front-end was managing the **real-time interaction** between the user interface and the backend database. To ensure efficient data handling, Firebase was integrated into the application. **Firebase Firestore** serves as the primary database, allowing users to store and retrieve their menstrual cycle details, symptoms, and health-related data in real-time. The **Firebase Authentication** module was implemented to handle user login and registration securely, using email-password authentication.

Various approaches were considered, including **setState(), Provider, and Riverpod**. Initially, **setState()** was used for simple UI updates, but as the project expanded, **Provider** was chosen for its efficiency in handling application-wide state changes. This ensured a **smooth data flow between the UI and Firestore**, allowing users to see updates instantly when they log or edit their health information.

The integration of **Flutter and Firebase** significantly improved the efficiency of the HER AURA menstrual tracker. With a well-structured UI, real-time data synchronization, and a robust authentication system, the app provides a **secure, intuitive, and effective** way for users to track their menstrual cycles and manage their reproductive health. The outputs of the same are shared in the Results and discussion session (4).

3.6 ISSUES AND FIXES

During the development of the HER AURA menstrual tracker app, several challenges were encountered that required troubleshooting and optimizations. The most common issues were related to **Firestore integration, data synchronization, state management, and app crashes**. Each problem was addressed systematically to ensure the app functions smoothly and efficiently.

3.6.1 Firebase Authentication Errors

One of the initial problems was **authentication failures** during user login and registration. Firebase Authentication was used to allow users to sign up and log in with their email and password. However, errors occurred due to incorrect **Firebase configuration settings**. The issue was resolved by:

- Properly configuring **Google Services JSON** for Android and **Google Info.plist** for iOS.
- Enabling **Email/Password Authentication** in Firebase settings.
- Implementing **error handling mechanisms** to display proper messages for failed authentication attempts.

3.6.2. Data Sync Delays in Firestore

At times, user data did not update immediately in the application, causing inconsistencies in the displayed information. The problem was due to inefficient Firestore queries and improper use of **asynchronous functions**. The following optimizations were made:

- **Using StreamBuilder:** Implemented Firestore **StreamBuilder** in Flutter to listen for real-time data updates instead of manually fetching data on every action.
- **Optimized Firestore Queries:** Indexed the database correctly and used efficient query methods to **reduce read latency**.
- **Handled Asynchronous Functions Properly:** Used **async/await** for Firestore operations to prevent race conditions.

3.6.3. State Management Issues

In the early stages of development, the app experienced **UI inconsistencies and lag** due to improper state management. Initially, **setState()** was used, but it became inefficient as the project grew. To resolve this:

- **Provider was implemented** for efficient state management across multiple widgets.
- **Listeners and Change Notifiers** were used to update UI elements dynamically without excessive rebuilds.
- **Lazy Loading Techniques** were introduced to improve app performance and memory efficiency.

3.6.4. App Crashes and Debugging

Several crashes occurred due to **null values, improper API calls, and incorrect data types**. These crashes were fixed by:

- **Implementing Null Safety:** Ensured all variables were properly initialized to prevent null reference errors.
- **Using Try-Catch Blocks:** Wrapped Firebase API calls in **try-catch** blocks to handle unexpected failures.
- **Debugging with Flutter DevTools and Logcat:** Identified and fixed runtime errors using logging tools.

By addressing these issues systematically, the HER AURA app was significantly optimized for **performance, accuracy, and user experience**. An Example Issue is attached Below.

```
java.lang.NullPointerException: Attempt to invoke virtual method 'void android.view.Display.getSize(android.graphics.Point)' on a null object reference
    at org.chromium.ui.display.DisplayAndroid.updateFromDisplay(DisplayAndroid.java:280)
    at org.chromium.ui.display.DisplayAndroidManager$DisplayListenerBackendImpl.onDisplayChanged(DisplayAndroidManager.java:176)
    at android.hardware.display.DisplayManagerGlobal$DisplayListenerDelegate.handleMessage(DisplayManagerGlobal.java:786)
    at android.os.Handler.dispatchMessage(Handler.java:102)
    at android.os.Looper.loop(Looper.java:148)
    at android.app.ActivityThread.main(ActivityThread.java:7406)
```

Figure 3.5 Null pointer Exception Error

4. RESULTS AND DISCUSSION

The HER AURA Menstrual Tracker was developed to provide users with an intuitive and data-driven menstrual health tracking experience. The application successfully enables users to log their cycle data, track symptoms, receive food recommendations, and access pregnancy-related nutritional guidance. The results obtained from the implementation are analyzed based on several key parameters: cycle tracking accuracy, symptom-based recommendations, usability, efficiency.

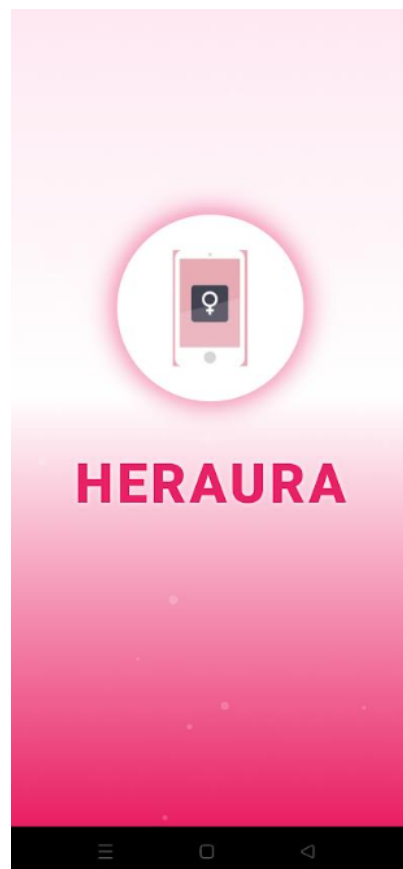


Figure 4.1 Splash screen

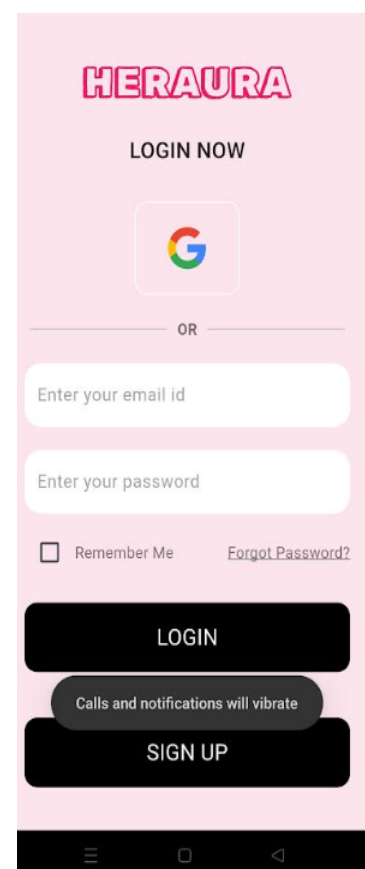


Figure 4.2 Login Screen

The **Splash Screen** is the first screen users see when they open the HER AURA app. It serves as an introduction to the application and helps establish branding.

The **Login Screen** is where users authenticate themselves before accessing the application. Since menstrual health data is sensitive, security is a priority.

Figure 4.3 SignUp Screen

Figure 4.4 Profile

The **Signup Screen** allows new users to create an account and set up their profile for personalized menstrual tracking.

The **Profile Page** in the HER AURA app plays a vital role in personalizing the user experience and managing essential health-related data.

It allows users to view, edit, and update their personal details and menstrual cycle information, ensuring accurate tracking and recommendations.

The page is designed for **ease of access, customization, and data security**, making it a crucial feature for maintaining an efficient menstrual tracking system.

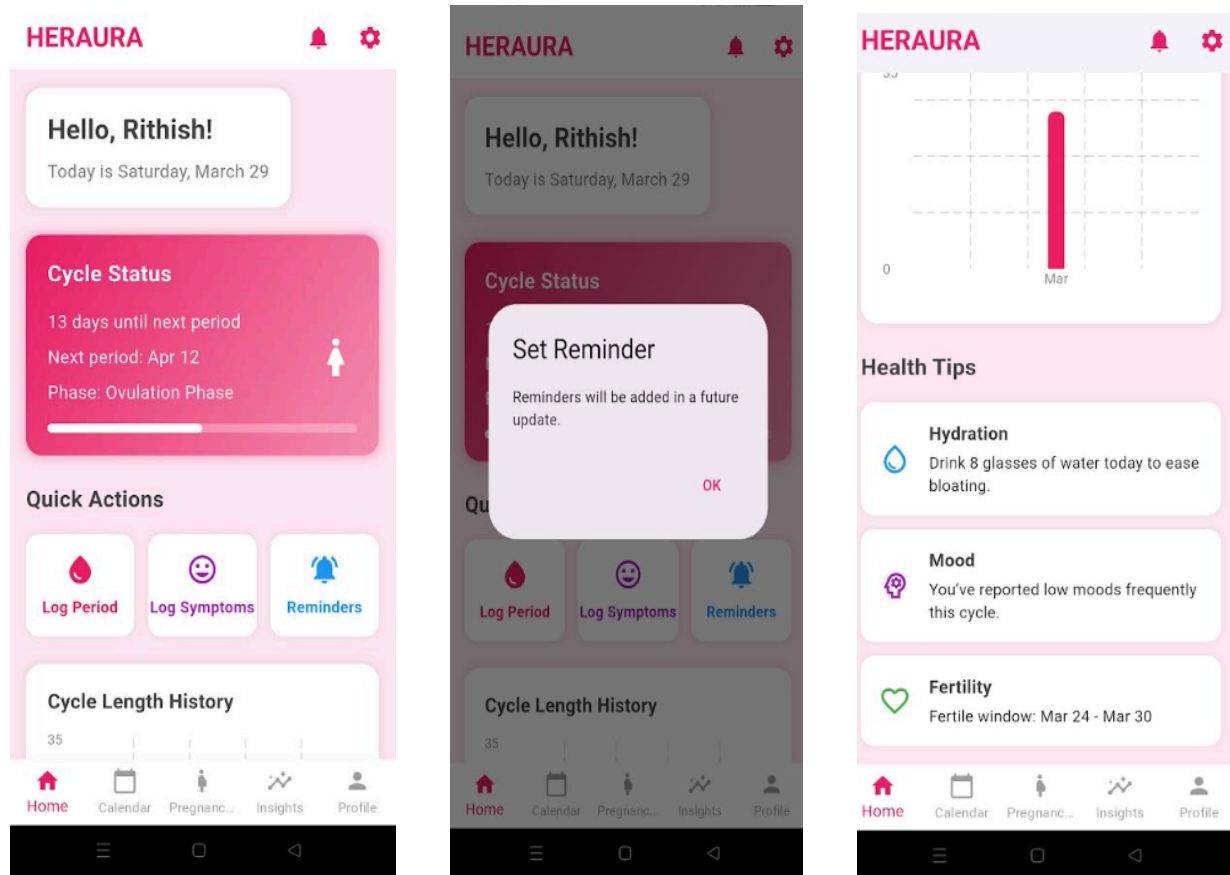


Figure 4.5a ,4.5 b, 4.5c Home Screen

The **Home Screen** of the HER AURA app serves as the central hub for users to access all essential menstrual tracking features, health insights, and personalized recommendations. Designed with an intuitive UI, it provides a seamless user experience, displaying key menstrual cycle information, symptom tracking options, and health recommendations in a visually appealing and easy-to-navigate manner.

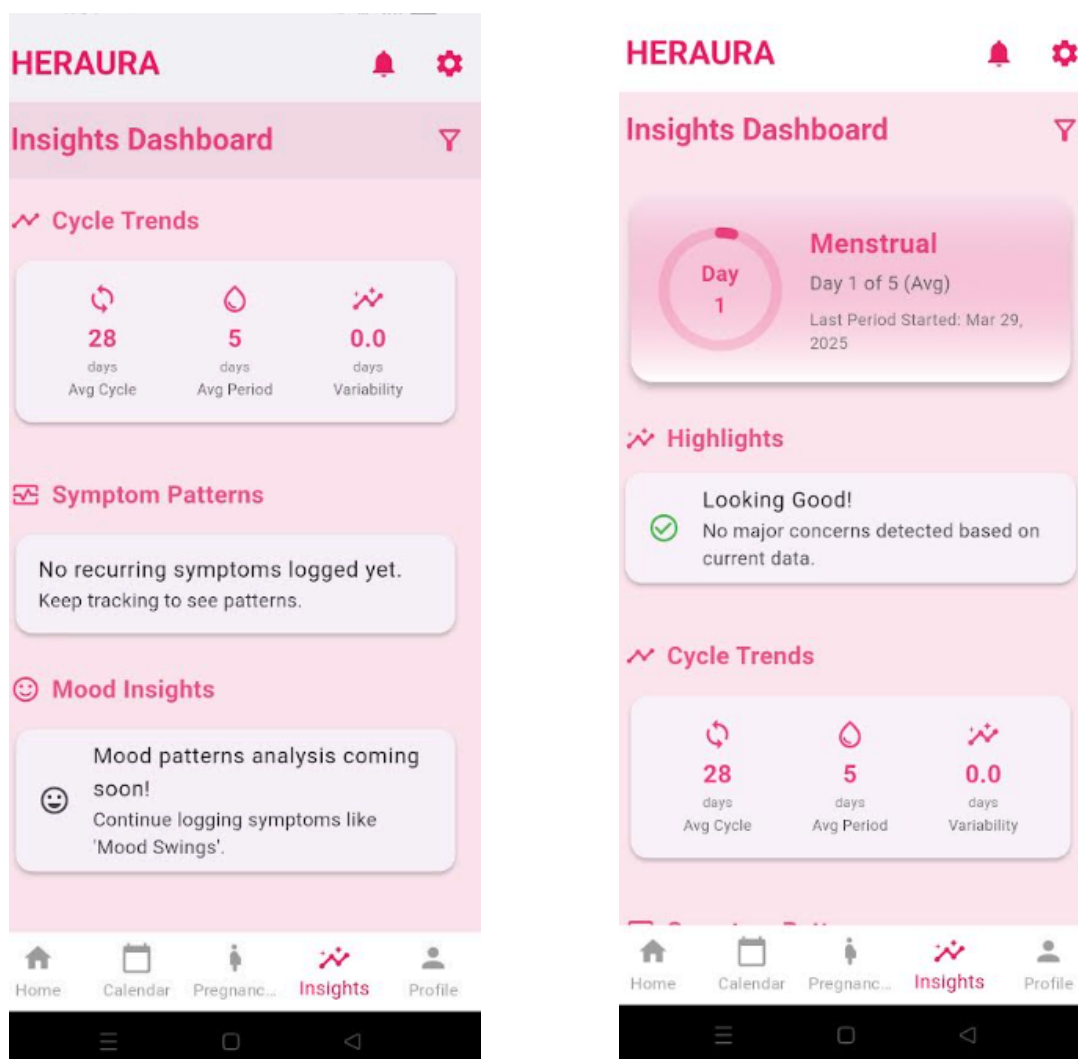


Figure 4.6a, 4.6b Insights Page

The Insights Page in the HER AURA app is a crucial feature that provides users with personalized menstrual health analytics, symptom trends, and wellness insights based on their logged data. It enables users to track patterns in their menstrual cycle, understand recurring symptoms, and receive tailored recommendations for maintaining optimal health.

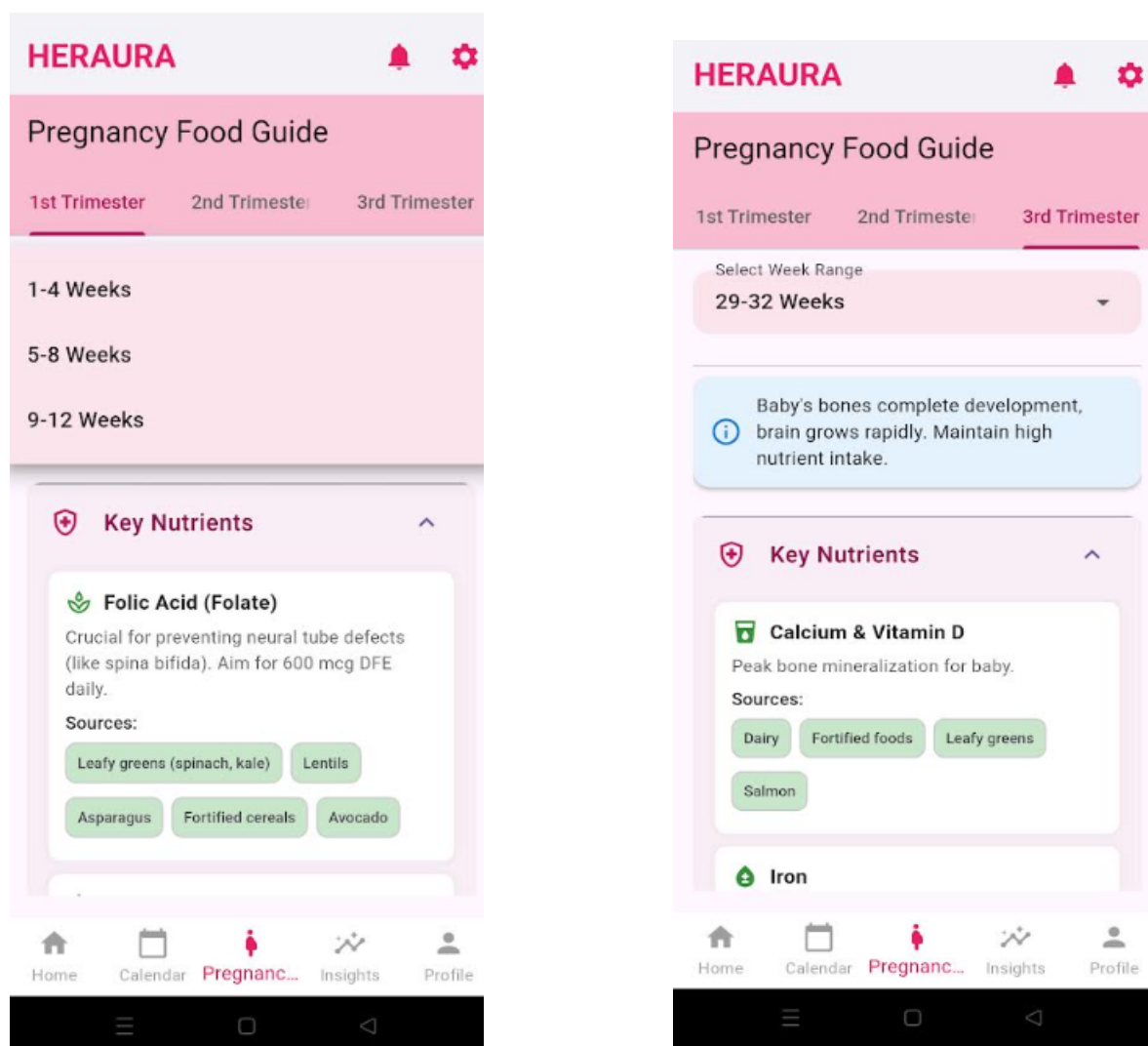


Figure 4.7a, 4.7b Pregnancy Guide

Designed to support expectant mothers with essential nutritional information throughout their pregnancy journey. It highlights key developmental milestones of the baby and provides guidance on maintaining a healthy diet to meet increased nutrient needs. The page lists vital nutrients along with their benefits and food sources, making it easy for users to incorporate them into their daily meals. With a user-friendly interface and clear recommendations, this guide empowers mothers to make informed choices for a **healthy pregnancy**.

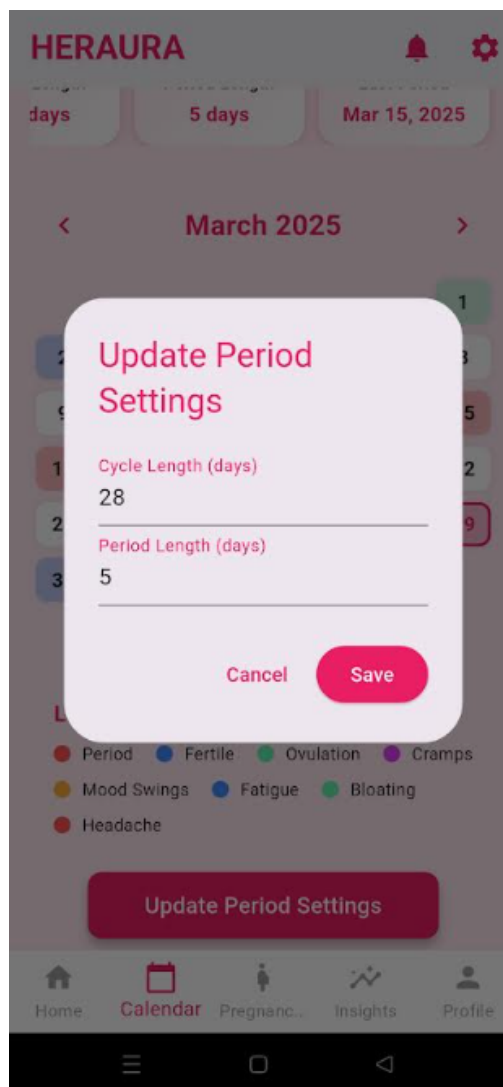
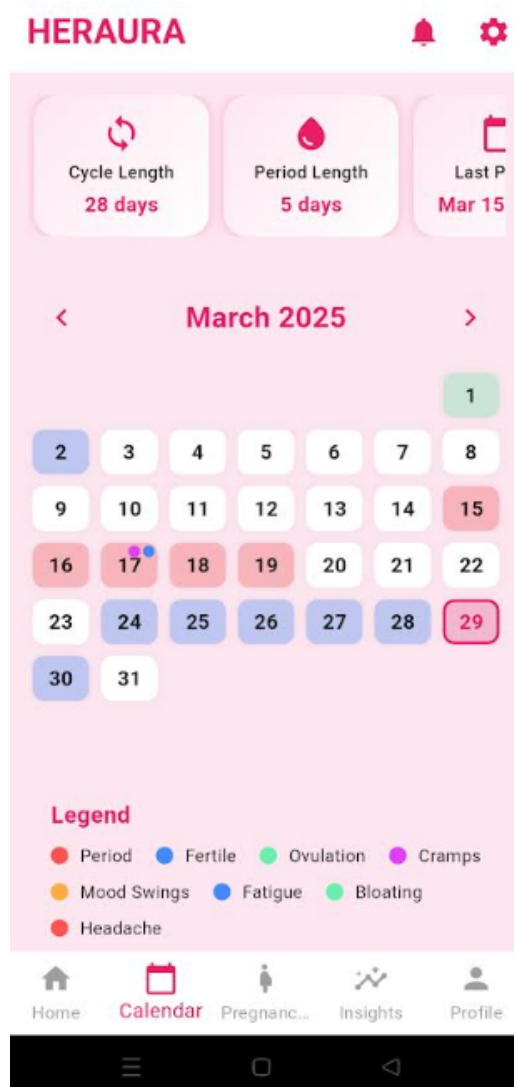


Figure 4.8a, 4.8b. Menstrual Calendar

The HERAURA app offers a streamlined menstrual cycle tracking experience with a user-friendly calendar interface. It allows users to monitor their cycle length, period duration, and log symptoms like cramps, fatigue, and mood swings through an easy-to-read legend. The calendar provides predictions for upcoming periods, fertile windows, and ovulation dates based on logged data. With simple navigation tabs for home, calendar, pregnancy insights, and profile, the app combines functionality with clarity.

5.CONCLUSION AND FUTURE ENHANCEMENT

5.1 CONCLUSION

The HER AURA menstrual tracker successfully provides a comprehensive and user-friendly solution for monitoring menstrual health, tracking symptoms, and offering personalized wellness recommendations. By enabling users to log their cycle details, symptoms, and overall well-being, the app helps them gain better awareness and control over their reproductive health. The food suggestions based on symptoms and pregnancy-week-specific dietary guidance add a valuable health-supporting feature that enhances the overall user experience.

The application ensures accurate period predictions, timely reminders, and insightful health analytics, helping users make informed decisions about their daily routines. The structured tracking system allows individuals to identify patterns in their cycles, anticipate potential discomfort, and take proactive measures to improve their well-being. The ability to provide personalized health insights and recommendations ensures that users receive tailored guidance rather than generic information.

Overall, the app proves to be an effective and supportive tool for individuals seeking to manage their menstrual and reproductive health efficiently. It not only simplifies period tracking but also empowers users by offering meaningful health insights and lifestyle recommendations.

5.2 FUTURE ENHANCEMENT

To further improve the app's impact and usability, several enhancements can be made:

- **AI-Powered Insights** – Using machine learning for better symptom prediction and personalized recommendations.
- **Wearable Device Integration** – Syncing with smartwatches and fitness trackers for real-time health monitoring.
- **Telehealth Support** – Allowing users to consult with doctors and nutritionists directly through the app.

- **Community & Support Forum** – A space where users can share experiences and seek advice.
- **Multi-Language and Accessibility Features** – Ensuring inclusivity by supporting different languages and assistive technologies.

With these advancements, HER AURA can evolve into a **more intelligent and holistic menstrual health companion**, offering users greater insights, better health management tools, and enhanced personalization.

REFERENCES

1. Ko, S., Lee, J., An, D., & Woo, H. (2023). Menstrual tracking mobile app review by consumers and health care providers: Quality evaluations study. *JMIR mHealth and uHealth*, 11(1), e40921.
2. Grieger, J. A., & Norman, R. J. (2020). Menstrual cycle length and patterns in a global cohort of women using a mobile phone app: Retrospective cohort study. *Journal of Medical Internet Research*, 22(6), e17109.
3. Symul, L., Wac, K., Hillard, P., & Salathé, M. (2019). Assessment of menstrual health status and evolution through mobile apps for fertility awareness. *NPJ Digital Medicine*, 2(1), 64.
4. Trépanier, L. C. M., Lamoureux, É., Bjornson, S. E., Mackie, C., Alberts, N. M., & Gagnon, M. M. (2023). Smartphone apps for menstrual pain and symptom management: A scoping review. *Internet Interventions*, 31, 100605.
5. Stanford, J. B., Willis, S. K., Hatch, E. E., Rothman, K. J., & Wise, L. A. (2020). Fecundability in relation to use of mobile computing apps to track the menstrual cycle. *Human Reproduction*, 35(10), 2245-2252.
6. Lee, Y., & Moon, M. (2016). Utilization and content evaluation of mobile applications for pregnancy, birth, and child care. *Healthcare Informatics Research*, 22(2), 73-80.
7. Doherty, K., Barry, M., Marcano-Belisario, J., Arnaud, B., Morrison, C., Car, J., & Doherty, G. (2018). A mobile app for the self-report of psychological well-being during pregnancy (BrightSelf): Qualitative design study. *JMIR Mental Health*, 5(4), e10007.
8. Overdijkink, S. B., Velu, A. V., Rosman, A. N., Van Beukering, M. D. M., Kok, M., & Steegers-Theunissen, R. P. M. (2018). The usability and effectiveness of mobile health technology-based lifestyle and medical intervention apps supporting health care during pregnancy: Systematic review. *JMIR mHealth and uHealth*, 6(4), e8834.

9. Carter, J., Sandall, J., Shennan, A. H., & Tribe, R. M. (2019). Mobile phone apps for clinical decision support in pregnancy: A scoping review. *BMC Medical Informatics and Decision Making*, 19, 1-13.
10. Musgrave, L. M., Kizirian, N. V., Homer, C. S. E., & Gordon, A. (2020). Mobile phone apps in Australia for improving pregnancy outcomes: Systematic search on app stores. *JMIR mHealth and uHealth*, 8(11), e22340.
11. Vidal, C. (2024). Ethical considerations in menstrual tracking. *Nature Human Behaviour*.
12. Hong, M., Rajaguru, V., Kim, K., Jang, S., & Lee, S. (2024). Menstrual cycle management and period tracker app use in millennial and Generation Z individuals: Mixed methods study. *Journal of Medical Internet Research*, 26, e53146. <https://doi.org/10.2196/53146>
13. Schantz, J. S., Fernandez, C. S. P., & Anne Marie, Z. J. (2021). Menstrual cycle tracking applications and the potential for epidemiological research: A comprehensive review of the literature. *Current Epidemiology Reports*, 8(1), 9-19. <https://doi.org/10.1007/s40471-020-00260-3>
14. Usala, T., et al. (2024). Comparison of hormonal levels measured by MiraTM Monitor and serum assays in ovulatory cycles. *Medicina (Kaunas)*.
15. Karlsson, M. (2023). Reimagining the cycle: Interaction in self-tracking period apps and menstrual empowerment. *Frontiers in Computer Science*, 5, 1166210.